

(1.1) Dokažte, že v libovolném komplexním vektorovém prostoru V se skalárním součinem $\langle \cdot, \cdot \rangle$ platí (pro libovolné $u, v \in V$)

$$1) \quad \operatorname{Re}(\langle u, v \rangle) = \frac{1}{4}(\|u + v\|^2 - \|u - v\|^2),$$

$$2) \quad \operatorname{Im}(\langle u, v \rangle) = \frac{1}{4}(\|u - iv\|^2 - \|u + iv\|^2),$$

kde $\operatorname{Re}(x)$ značí reálnou a $\operatorname{Im}(x)$ imaginární část komplexního čísla x . (Pro inspiraci vizte tvrzení 8.32 ve skriptech.)

$$\begin{aligned} 1) \quad \|u + v\|^2 &\stackrel{\text{def.}}{=} \langle u+v, u+v \rangle \stackrel{(SL2)}{=} \underbrace{\langle u, u \rangle}_{\|u\|^2} + \underbrace{\langle u, v \rangle + \langle v, u \rangle}_{2 \operatorname{Re}(\langle u, v \rangle)} + \underbrace{\langle v, v \rangle}_{\|v\|^2} = \\ &= \|u\|^2 + \underbrace{\langle u, v \rangle + \langle v, u \rangle}_{2 \operatorname{Re}(\langle u, v \rangle)} + \|v\|^2 = \|u\|^2 + 2 \operatorname{Re}(\langle u, v \rangle) + \|v\|^2 \end{aligned}$$

$$\begin{aligned} \|u - v\|^2 &\stackrel{\text{def.}}{=} \langle u-v, u-v \rangle \stackrel{(SL2)}{=} \underbrace{\langle u, u \rangle}_{\|u\|^2} - \underbrace{\langle u, v \rangle + \langle v, u \rangle}_{2 \operatorname{Re}(\langle u, v \rangle)} + \underbrace{\langle v, v \rangle}_{\|v\|^2} = \\ &= \|u\|^2 - \underbrace{\langle u, v \rangle + \langle v, u \rangle}_{2 \operatorname{Re}(\langle u, v \rangle)} + \|v\|^2 = \|u\|^2 - 2 \operatorname{Re}(\langle u, v \rangle) + \|v\|^2 \end{aligned}$$

$$\operatorname{Re}(\langle u, v \rangle) = \frac{1}{4}(\|u + v\|^2 - \|u - v\|^2) = \frac{1}{4}(4 \operatorname{Re}(\langle u, v \rangle)) = \operatorname{Re}(\langle u, v \rangle)$$

$$\begin{aligned} 2) \quad \|u - iv\|^2 &\stackrel{\text{def.}}{=} \langle u-iv, u-iv \rangle = \underbrace{\langle u, u \rangle}_{\|u\|^2} + \underbrace{\langle u, -iv \rangle}_{(-i)\langle u, v \rangle} + \underbrace{\langle -iv, u \rangle}_{(-i)\langle v, u \rangle} + \underbrace{\langle -iv, -iv \rangle}_{\|v\|^2} = \\ &= \|u\|^2 - i\langle u, v \rangle + i\overline{\langle u, v \rangle} + \|v\|^2 = \|u\|^2 - i(\langle u, v \rangle - \overline{\langle u, v \rangle}) + \|v\|^2 = \|u\|^2 - 2i \operatorname{Im}(\langle u, v \rangle) + \|v\|^2 = \\ &= \|u\|^2 + 2 \operatorname{Im}(\langle u, v \rangle) + \|v\|^2 \end{aligned}$$

$$\begin{aligned}
 \underline{\|u + ir\|^2} &\stackrel{\text{def.}}{=} \langle u + ir, u + ir \rangle = \underbrace{\langle u, u \rangle}_{\substack{\text{def.} \\ \|u\|^2}} + \underbrace{\langle u, ir \rangle}_{\substack{\|(\cdot, \cdot)\| \\ i \langle u, r \rangle}} + \underbrace{\langle ir, u \rangle}_{\substack{\|(\cdot, \cdot)\| \text{ sss} \\ \overline{i \langle r, u \rangle}}} + \underbrace{\langle ir, ir \rangle}_{\substack{\|(\cdot, \cdot)\| \text{ sss} \\ \overline{i \cdot i \langle r, r \rangle} \\ \stackrel{\text{def.}}{\|r\|^2}}} = \\
 &= \|u\|^2 + i \underbrace{\left(\langle u, r \rangle - \overline{\langle u, r \rangle} \right)}_{2i \operatorname{Im}(\langle u, r \rangle)} + \|r\|^2 = \|u\|^2 + 2i^2 \operatorname{Im}(\langle u, r \rangle) + \|r\|^2 = \\
 &= \underline{\|u\|^2 - 2 \operatorname{Im}(\langle u, r \rangle) + \|r\|^2}
 \end{aligned}$$

$$\operatorname{Im}(\langle u, r \rangle) = \frac{1}{4} (\|u - ir\|^2 - \|u + ir\|^2) = \frac{1}{4} (4 \operatorname{Im}(\langle u, r \rangle)) = \operatorname{Im}(\langle u, r \rangle)$$

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